

PROGRAM 7: Deploy a web application to kubernetes using minikube

PROJECT STRUCTURE :

.....> nginx-deployment.yaml

>mkdir program7

>cd program7

Step1: Start minikube

Minikube start

Minikube status

```
Jan 5 12:14
Terminal
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~$ cd program7
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ ls
nginx-deployment.yaml
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ minikube start
🐹 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
🚀 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌞 Enabled addons: default-storageclass, storage-provisioner, dashboard
🏡 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$
```

Step2: create a nginx deployment.yaml and add the following code

```
GNU nano 7.2 nginx
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step3: Apply the nginx-deployment.yaml

>kubectl apply -f nginx-deployment.yaml

Check the status

>kubectl get deployments

>Kubectl get pods

```

1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ nano nginx-deployment.yaml
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment configured
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    3/3     3             3           6d23h
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5j8rb   1/1     Running   3 (4d21h ago)   6d23h
nginx-deployment-6f9664446b-hzjrm   1/1     Running   3 (4d21h ago)   6d23h
nginx-deployment-6f9664446b-w9npc   1/1     Running   0           26s
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ 

```

Step4: expose the deployment as a service

```

>Kubectl expose deployment nginx-deployment \
> type=Nodeport -/
>port=80

```

Step5: check service

```

>Kubectl get services

```

Step6:access the web app

```

>minikube service nginx-deployment

```

```

see 'kubectl expose --help' for usage.
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ kubectl expose deployment nginx-deployment
--type=NodePort \
--port=80
service/nginx-deployment exposed
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ kubectl get services
NAME            TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)        AGE
kubernetes      ClusterIP   10.96.0.1     <none>       443/TCP        18d
nginx-deployment NodePort    10.98.76.90   <none>       80:31590/TCP   96s
nginx-service   NodePort    10.111.247.118 <none>       80:32375/TCP   5d
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ minikube services nginx-deployment
Error: unknown command "services" for "minikube"

```

Did you mean this?
service

```

Run 'minikube --help' for usage.
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ minikube service nginx-deployment

```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31590

```

🌐 Opening service default/nginx-deployment in default browser...
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program7$ Opening in existing browser session.
]

```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.